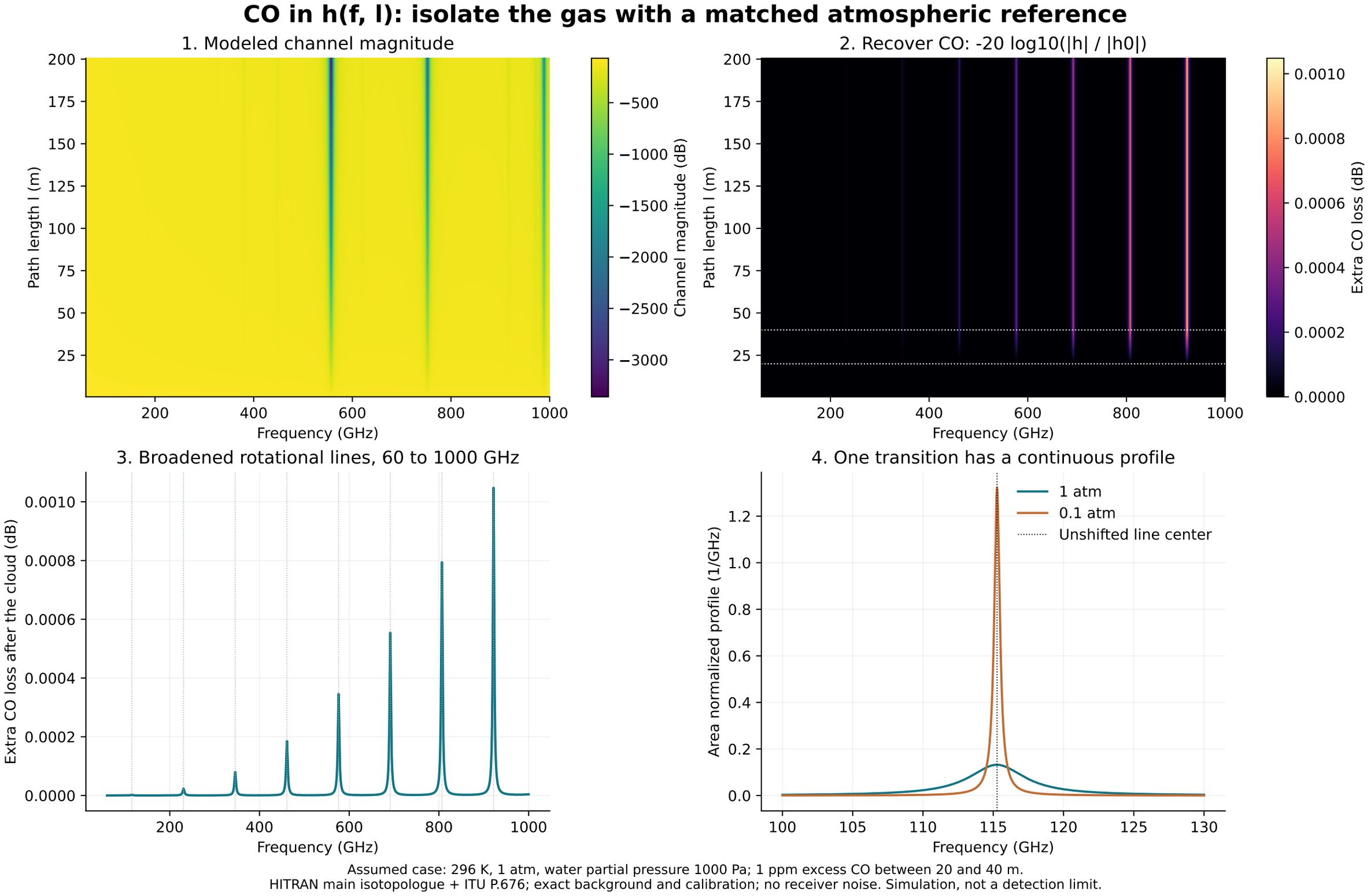
CO in h(f, l): isolate the gas with a matched atmospheric reference
1. Modeled channel magnitude
Path length l (m)
Frequency (GHz)
Channel magnitude (dB)
2. Recover CO: -20 log10(|h| / |h0|)
Path length l (m)
Frequency (GHz)
Extra CO loss (dB)
3. Broadened rotational lines, 60 to 1000 GHz
Extra CO loss after the cloud (dB)
Frequency (GHz)
4. One transition has a continuous profile
1 atm
0.1 atm
Unshifted line center
Area normalized profile (1/GHz)
Frequency (GHz)
Assumed case: 296 K, 1 atm, water partial pressure 1000 Pa; 1 ppm excess CO between 20 and 40 m.
HITRAN main isotopologue + ITU P.676; exact background and calibration; no receiver noise. Simulation, not a detection limit.